

three-wheelers and 15 four-wheelers) were sold in 2018-19 in the city.

There are hundred Indian cities that are a part of the smart cities mission. In Ahmedabad, the first fleet of electric buses along with the country's first automated battery charging and swapping station for these buses was inaugurated in 2019. The buses have automatic door sensors that prevent them from moving if doors

are open. Out of 550 electric buses planned for Gujarat under Faster Adoption and Manufacturing of (Hybrid &) Electric Vehicles (FAME) India scheme, the city alone has a share of 300 buses.

In a recent list released by Union Ministry of Housing and Urban Affairs, Nagpur and Ahmedabad were among the top three cities to implement smart city projects in 2019. Globally,

many cities, including Amsterdam, New York, Singapore, London, Seoul and Hong Kong, have already implemented numerous smart solutions at a large scale. By incorporating the required changes while replicating the EV model in other parts of the country, India can have a financially-viable, zero-emission, hassle-free transportation system in the future.

—Ayushee Sharma

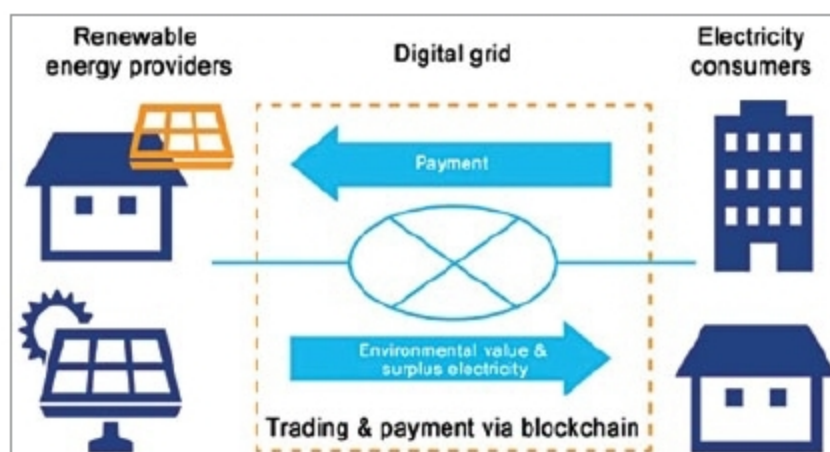
2 ROLE OF BLOCKCHAIN IN GREEN ENERGY

Besides doing away with the need for a centralised electric vehicle (EV) charging infrastructure, blockchain eases trading between consumers and charging stations or transport providers

Ever since the concept of cryptocurrencies became popular with the hype around Bitcoin, its underlying technology, blockchain, has found application in numerous other sectors such as healthcare, real estate, tourism and media, among others, for data maintenance and security.

Data in a blockchain is added in chronological order in the form of blocks, which ensures immutability. In addition to this, updates are made to the blockchain only after consensus among all peers in the network, which makes the database decentralised. New additions are linked to previous ones through cryptography, which ensures security. Data formats can also be standardised across different institutions. All these features promote transparent sharing of information and establish trust among its users.

In the renewable energy sector too, established companies as well as startups, such as Acciona Energy, Irene Energy, The Sun Exchange, Electrify Asia, Greeneum and Iberdrola, have realised the potential of blockchain. Several attempts are being made to utilise it as a tool for catalysing the transition to green energy. According to a recent report from Research and Markets, blockchain in the energy market was valued at US\$ 518.6 million in 2018



(Credit: www.fujikura.co.jp)

and is projected to grow at a CAGR of 54.09 per cent during the forecast period from 2019 to 2024.

A smart electricity grid employs renewable energy as the preferred choice of energy. Renewable energy acts as backup power when the central grid fails to balance supply and demand. As more people get renewable energy installations in their homes, the number of prosumers in the market who can sell the excess energy generated increases. For instance, solar power from rooftop panels attached to a computer can provide energy at a low cost during the day when sunlight is in abundance.

This decentralisation of energy systems can create stress on traditional electricity grids. Blockchain can stabilise the grid through peer-to-peer (P2P) transaction platforms that enable buying and selling of energy generated from even small-scale installations. Pando by Lo3 Energy is

one such platform that allows pooling of local distributed energy resources and optimises the grid at the community level.

By linking energy grids to blockchain and using smart meters, data sharing is possible in real time. Electricity generated and consumed is tracked and recorded in a tamper-proof way, which ensures accountability between sellers and buyers. Customer energy usage

patterns can be identified, which makes it easier to supply the right amount of energy and reduce inequality in both areas, with or without access to central grids. Consumers can make more informed decisions as they can compare supplies from different energy providers, which drives competition.

Small producers and new players can also get green certificates for trading automatically through blockchain, which improves chances of higher investment in this field. This promotes local consumption and boosts growth of clean energy into the mainstream while simultaneously reducing transmission losses and the need for expensive infrastructure.

Different initiatives are being taken in India as well. For example, Uttar Pradesh government has collaborated with Australia's Power Ledger to pilot solar energy trading through blockchain integrated with smart meter sys-